

Coecobrya cf. Tenebricosa Springtail Jumping on Different Substrates and Water Droplet Collision

Background

Coecobrya cf. Tenebricosa, known as Tropical Pink Springtails, are 1-2mm long, wingless hexapods which jump using furca, a bifurcated appendage which folds out from underneath the body in a hinge-like motion. Springtails have the remarkable ability to jump hundreds of times their body length^[2] while righting their body angle for controlled landing in less than ~20 milliseconds, the fastest in recorded animals^[1]. Understanding how springtail jumping changes across various substrates (land and water surfaces) and how they adapt to things like raindrop collisions and the resulting crown formations may later enhance development of comprehensive, agile jumping mechanisms in biomimetic robotics^[3].



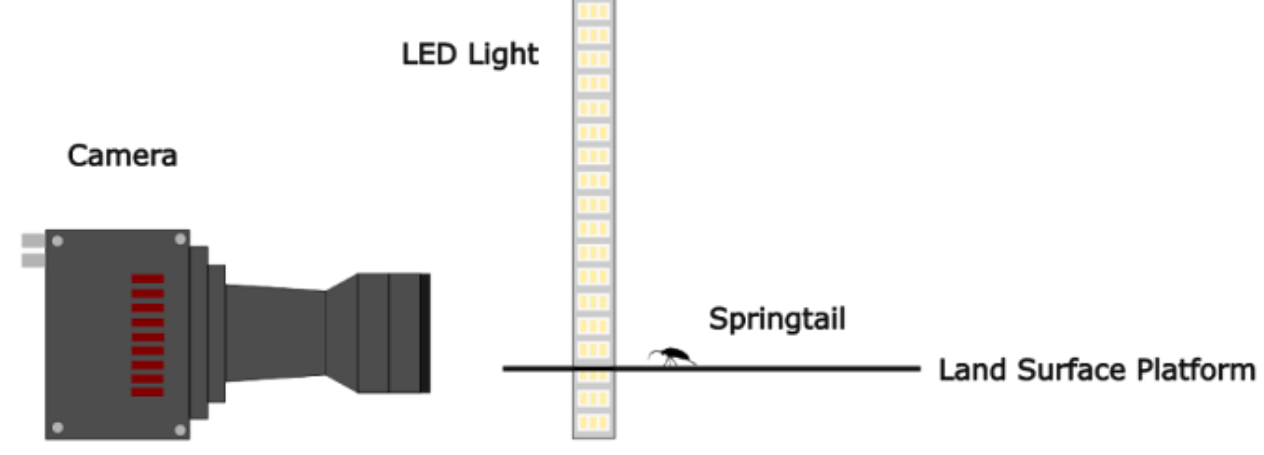
Fig 1. *Coecobrya cf. Tenebricosa* in color (top) and greyscale (bottom), enlarged to show detail of anatomy and furca appendage.

Setup & Method – Substrate Type

Analysis done using DLTdv8 MATLAB application and ImageJ's Kappa software.

The land surface was a 12x12-inch, black acrylic platform. The water surface was a 5x2.5x6-inch clear acrylic container with 200 mL of water. We observed one springtail at a time, illuminating with LED's.

On Land:



On Water:

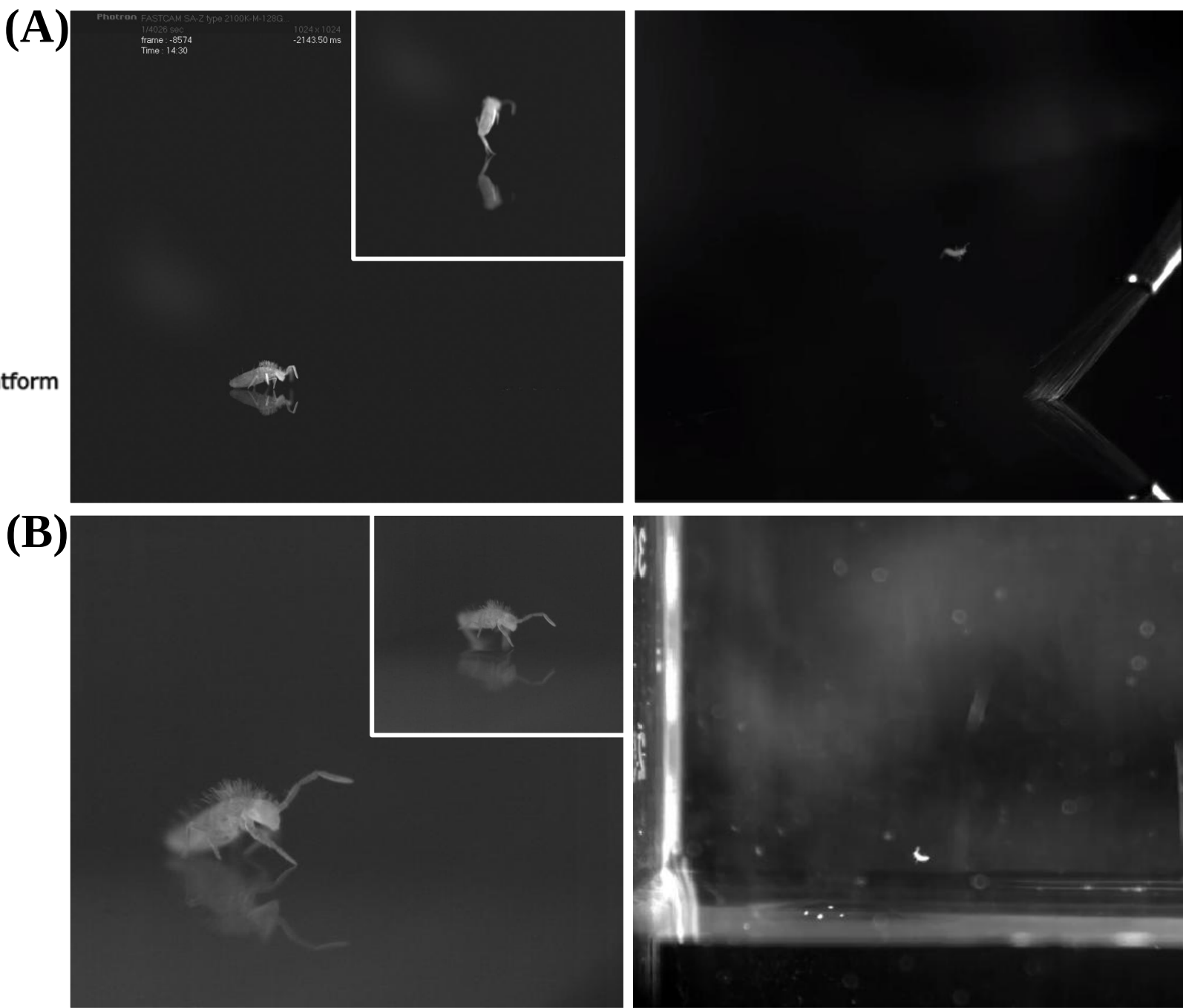
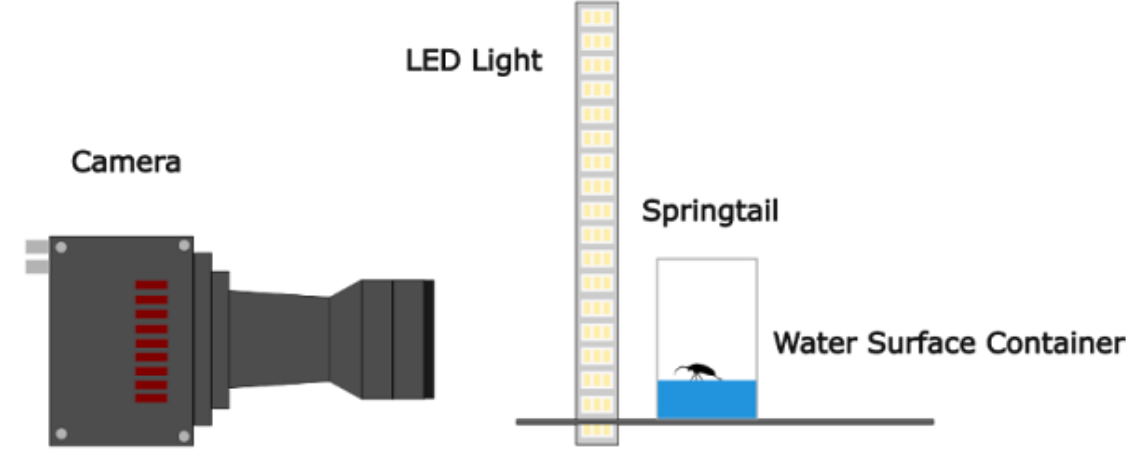


Fig 2. (Left) Schematic setup of the experimental design for surface type. On land and on water, the high speed camera observes jumping springtails. (Right) Experiment recording of detailed takeoffs and full trajectories for (A) on land and (B) on water.

Objectives & Questions

Understand how springtail jumping mechanisms differs between land and water jumping surfaces: **Do overall springtail jumping trajectories differ between land and water? Do springtail furca open differently when jumping on land versus water?** Understand springtail ejection from the water's surface after being hit by drops from varying heights: **Does springtail ejection trajectory change with drop height/velocity at the time of collision?**

Springtail Trajectory & Furca Opening

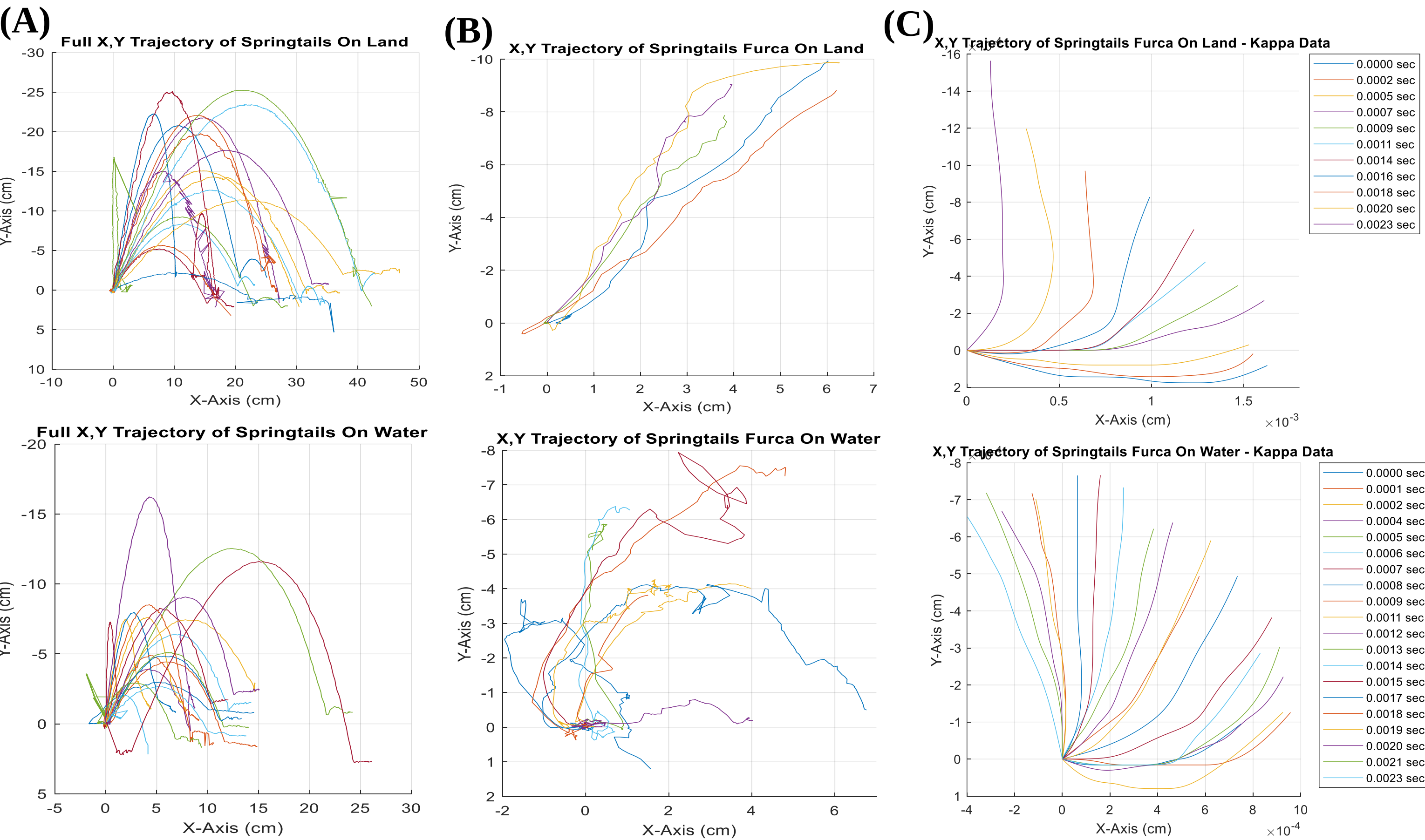


Fig 3. MATLAB plots of trajectories on land (above) and water (below) of (A) Full springtail jumping trajectories (B) Furca jumping trajectories and (C) Furca opening curvature

Setup & Method – Water Droplets

Analysis done using DLTdv8 MATLAB application.

We recorded the drop hitting the center of a mass of springtails congregated on the water's surface in a petri dish. We released 5 drops per height from a pipette raised to 30, 90, 120, and 150 cm.

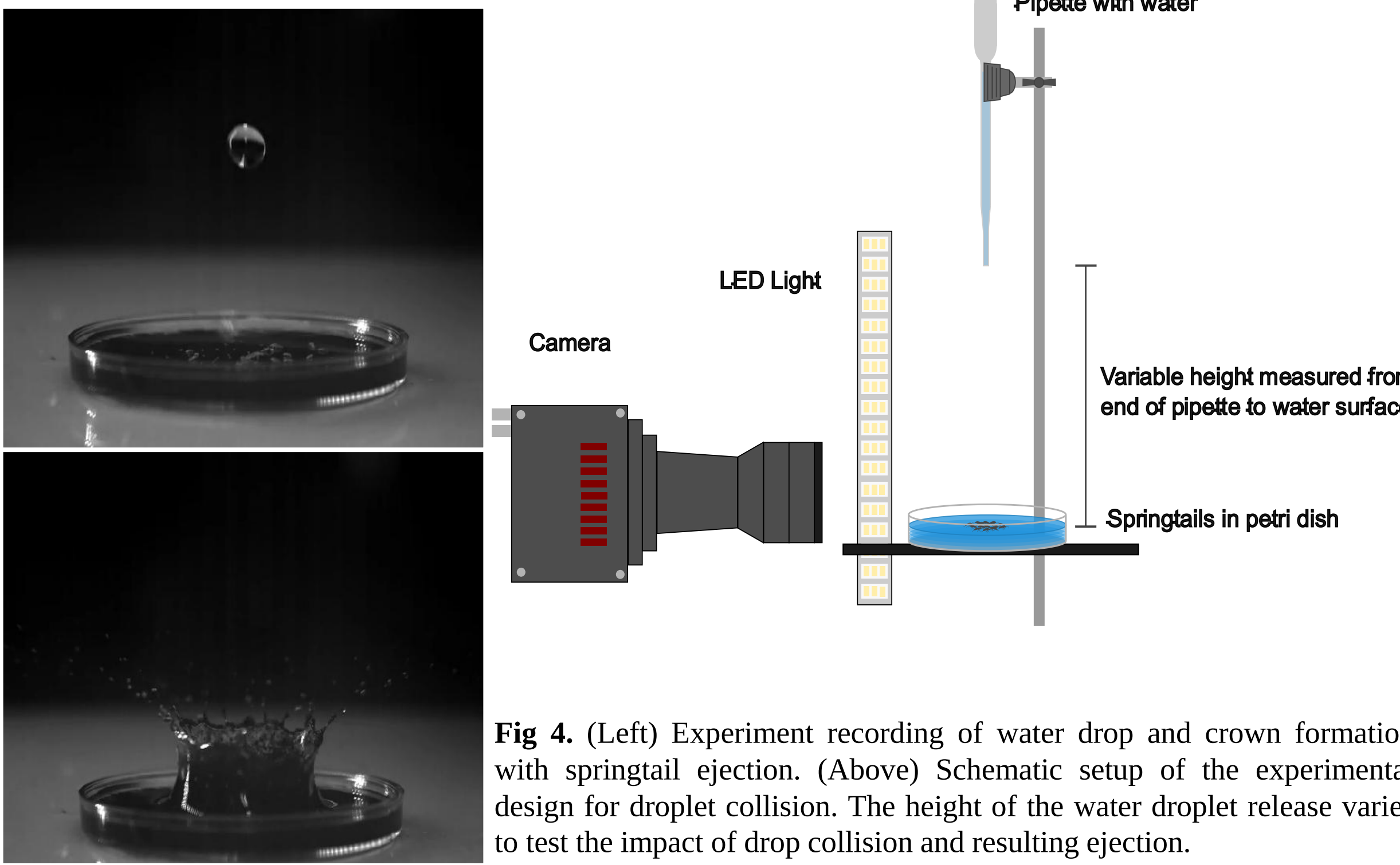


Fig 4. (Left) Experiment recording of water drop and crown formation with springtail ejection. (Above) Schematic setup of the experimental design for droplet collision. The height of the water droplet release varies to test the impact of drop collision and resulting ejection.

Summary

Substrate Type:

- Full jumping trajectories are higher and further on land compared to water
- On land, furca stay stationary while springtail bodies move backward into the jump trajectory
- On water, the furca open backwards simultaneously to the body moving in the jump trajectory

Water Drop Collision:

- Higher drop release height leads to greater drop velocity, generating a larger crown and splash radius
- Positive correlation between drop impact and springtail ejection distance and velocity

	Land	Water
Jump Duration (ms)	0.24 (0.15-0.66)	0.16 (0.10, 0.23)
Jump Height (cm)	1.32 (0.03-5.32)	0.61 (0.02-2.77)
Jump Distance (cm)	8.89 (0.01-39.3)	5.9 (0.01-26.07)
Takeoff Velocity (cm/s)	152.65 (5.94-211.4)	60.8 (3.87-191.2)

Table 1. Jump kinematics, mean (min-max), of *Coecobrya cf. Tenebricosa* on various substrates (land and water takeoff surfaces).

Drop Collision & Springtail Ejection

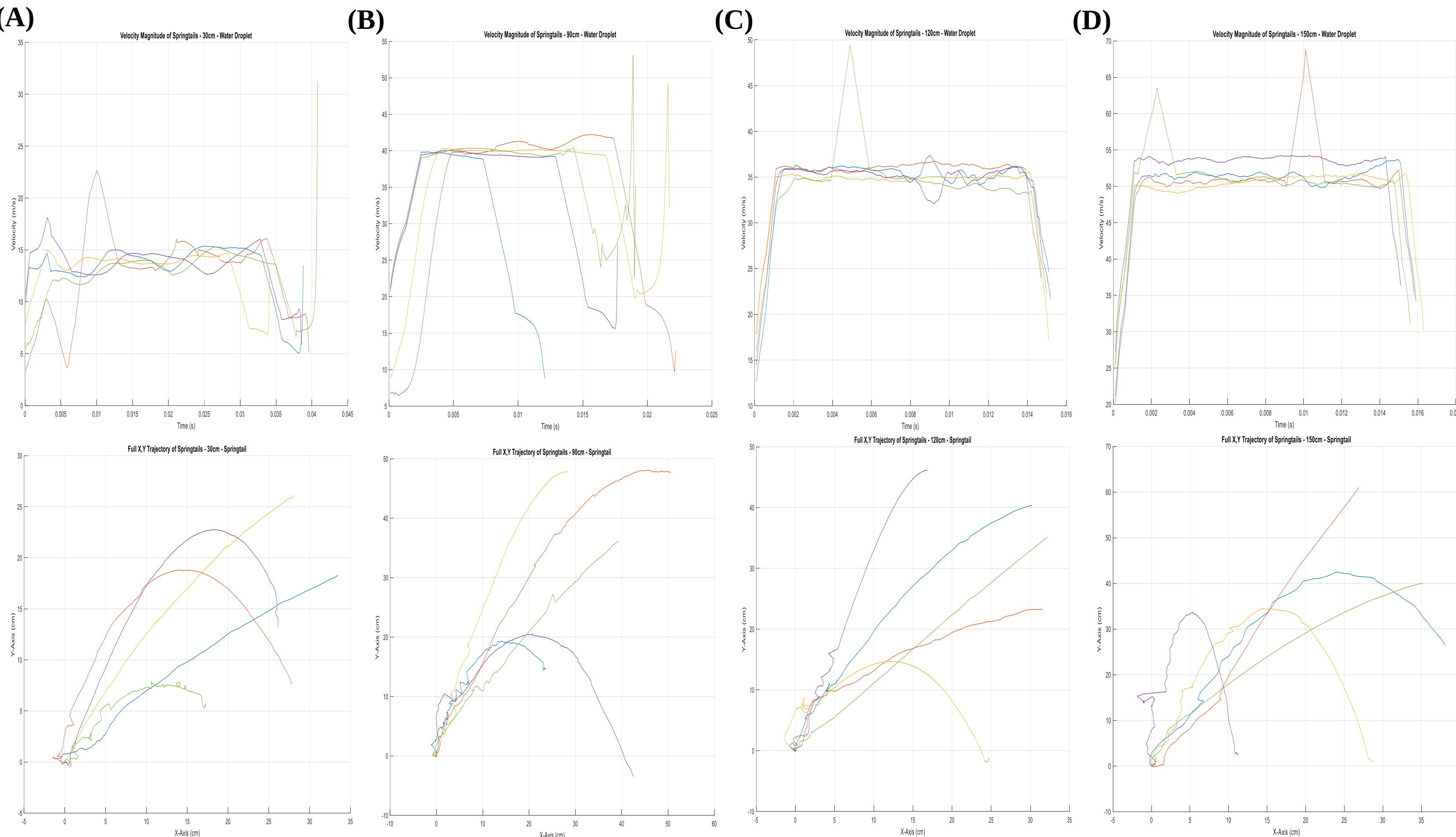


Fig 5. MATLAB plots of drop velocity (top) and springtail ejection trajectory (bottom) for (A) 30cm, (B) 90cm, (C) 120cm, and (D) 150cm droplet release heights.

References

- [1] Ortega-Jimenez, V. M., Challita, E. J., Kim, et al.. (2022). Directional takeoff, aerial righting, and adhesion landing of semiaquatic springtails. *Proceedings of the National Academy of Sciences of the United States of America*, 119(46), e2211283119. <https://doi.org/10.1073/pnas.2211283119>
- [2] A A Smith, J S Harrison, Jumping Performance and Behavior of the Globular Springtail *Dicyrtomina minuta*, *Integrative Organismal Biology*, Volume6, Issue1, 2024, obae029, <https://doi.org/10.1093/iob/obae029>
- [3] Francisco Ramirez Serrano et al., A springtail-inspired multimodal walking-jumping microrobot. *Sci.Robot.* **10**, eadp7854 (2025). DOI:[10.1126/scirobotics.adp7854](https://doi.org/10.1126/scirobotics.adp7854)